



Section 1: SQL — Databases & Queries

Tables & Data Types

1. What is a database? How is it different from a regular text file or spreadsheet?
2. What is a table in a relational database? What are rows and columns?
3. Write a SQL query to create a table named Students with columns: StudentId (INT, Primary Key), FirstName (NVARCHAR(50)), LastName (NVARCHAR(50)), Age (INT).
4. What is a Primary Key? Can a table have more than one Primary Key? Explain your answer.
5. What is the difference between INT, NVARCHAR, and DECIMAL data types in SQL? Give one example use case for each.
6. Write a SQL query to create a table named Courses with columns: CourseId (INT, Primary Key), CourseName (NVARCHAR(100)), DurationMonths (INT).
7. What is a Foreign Key? How does it connect two tables together? Give a real-world example.
8. Write a SQL query to create a table named Enrollments that links Students and Courses using foreign keys on both StudentId and CourseId.

INSERT, UPDATE & DELETE

9. Write SQL INSERT statements to add three student records into the Students table you created in Q3.
10. Write a SQL query to update the Age to 25 for the student where StudentId = 2.
11. Write a SQL query to delete the record from Students where StudentId = 5.
12. What happens if you try to delete a record in a parent table that is referenced by a Foreign Key in a child table? How can you handle this?
13. Write a SQL query to insert a new course into the Courses table with CourseName = 'Web Development' and DurationMonths = 6.

SELECT & Filtering

14. Write a SELECT query to retrieve all columns from the Students table.
15. Write a SELECT query to get only FirstName and LastName of all students.
16. Write a SELECT query to get all students whose Age is greater than 20.
17. Write a SELECT query to find all students whose FirstName starts with the letter 'J'. Use the LIKE keyword.
18. Write a SELECT query to count the total number of students in the Students table.

19. Write a SELECT query to get all students sorted by LastName in alphabetical order.

20. Write a SELECT query to get the top 3 students by Age (oldest first).

21. What is the WHERE clause? How is it different from HAVING? When would you use HAVING?

JOIN Queries

22. Explain what a JOIN does in SQL. Why do we need JOINS?

23. Write an INNER JOIN query to show each student's FirstName along with the CourseName they are enrolled in. Use the Students and Courses tables with the Enrollments table.

24. What is the difference between INNER JOIN and LEFT JOIN? Give an example scenario for each.

25. Write a LEFT JOIN query to list all courses and show any students enrolled. Courses with no students should still appear in the result.

26. Write a query using an INNER JOIN between two tables: Employees (EmpId, FullName, DepartmentId) and Departments (DepartmentId, DepartmentName). Show each employee's full name and their department name.

27. What is a self-join? Give an example of when you might need to join a table to itself.

Section 2: C# Programming Basics

Variables, Data Types & Methods

28. Write C# statements to declare variables of type int, string, bool, and double. Assign a value to each.

29. What is the difference between a value type and a reference type in C#? Give two examples of each.

30. Write a C# method named AddNumbers that takes two int parameters and returns their sum.

31. What is the difference between a void method and a method that returns a value? Write one example of each.

32. Write a C# if-else statement that checks if a number is positive, negative, or zero and prints an appropriate message.

33. Write a C# for loop that prints every even number between 1 and 20.

34. Write a C# while loop that keeps asking the user to enter a number until they enter 0.

35. Write a C# nested for loop that prints the following star pattern: row 1 has one star, row 2 has two stars, continuing up to row 5. Do not hardcode the pattern.

Arrays & Collections

36. Write a C# statement to declare and initialise a string array named fruits with five values of your choice.

- 37.** Write a C# for loop to print every element of an integer array named scores.
- 38.** Write a C# method that takes an integer array and returns the largest number in it.
- 39.** What is the difference between an array and a List<T> in C#? When is it better to use a List?
- 40.** Write C# code to create a List<string> named names, add three names to it, and then loop through and print each one.
- 41.** Write a C# method that takes a string array and returns the number of elements that contain the letter 'a'.

Section 3: Object-Oriented Programming (OOP)

- 42.** What is Object-Oriented Programming? Name the four main principles and give a one-sentence description of each.
- 43.** What is a Class? What is an Object? Write a C# class named Car with properties Make (string), Model (string), and Year (int). Then create an object of that class and set values for each property.
- 44.** What is a Constructor in C#? Rewrite the Car class from Q43 to include a constructor that sets all three properties when the object is created.
- 45.** Explain Encapsulation. Rewrite the Car class so that all properties are private and can only be accessed through public getters and setters.
- 46.** Explain Inheritance with a real-world example. Write a C# class named Vehicle with a method Describe() that prints 'This is a vehicle.' Then write a class named Car that inherits from Vehicle and overrides Describe() to print 'This is a car.'
- 47.** What is Polymorphism? Write an example in C# where a parent class reference variable holds a child class object and calls an overridden method.
- 48.** What is Abstraction? What is the difference between an abstract class and an interface in C#? Give one example of each.
- 49.** Write a C# interface named IShape with a method GetArea() that returns a double. Then write two classes, Circle and Rectangle, that implement this interface.

Section 4: .NET Minimal API

- 50.** What is a .NET Minimal API? How is it different from a regular ASP.NET MVC project?
- 51.** Write the C# code for a Minimal API endpoint that responds to GET /api/message and returns the JSON: { "message": "Hello from API" }.
- 52.** What is a connection string? Write an example connection string for a local SQL Server database named SchoolDB.
- 53.** Write a C# method named GetSymbol() that connects to a SQL database, reads the SymbolValue from a table named Symbols where Id = 1, and returns it as a string.

54. What is CORS? Why does a browser block API calls when CORS is not configured?
How do you enable CORS in a .NET Minimal API?

55. What is Swagger? How can you use it to test your API endpoint without writing any frontend code?

56. What is the difference between a GET request and a POST request? When would you use each one?

57. Write a C# Minimal API endpoint for POST `/api/symbol` that reads a symbol value from the request body and updates it in the database.

Section 5: HTML & JavaScript

58. Write an HTML page that has a heading saying Pattern Generator, a button with the text Generate Pattern, and a pre element with id output to display results.

59. Write a JavaScript function that uses `fetch()` to call the endpoint GET `/api/symbol`. Extract the symbol value from the JSON response and store it in a variable. Handle any errors by displaying an error message in the output element.

60. Using the symbol variable returned from the API call in Q59, write JavaScript code with nested for loops to build a triangle pattern and display it inside the pre element with id output. The number of rows should come from a user input, not be hardcoded. Explain each part of your code briefly.